Electric Stove



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Electric Stove

What it is used for

An electric stove converts electrical energy into heat for cooking or heating food in suitable cookware. Correct cookware selection, controlled heat settings, and safe electrical handling help ensure efficient cooking while reducing burn, shock, and fire hazards.

Main parts

- Heating element or hot plate
- Cooking surface or burner assembly
- Temperature and power controls
- Indicator light or display
- Power cord and plug or fixed electrical connection
- Housing, feet, and protective thermal components

Before you start

1. Confirm that the stove is designed for the available electrical supply and is placed on a stable, level, heat-resistant surface.
2. Inspect the heating surface, controls, power cord, plug, and housing for cracks, exposed wires, or other damage.
3. Ensure the surrounding area is dry, well ventilated, and clear of combustible materials.
4. Select cookware suitable for the stove type, with a flat base and an appropriate size for the heating zone.
5. Check that cookware handles are secure and positioned away from walkways and hot surfaces.
6. Keep a suitable lid, heat-resistant gloves, and appropriate fire safety equipment accessible.

How to use

1. Place the stove on a stable, level surface and keep its ventilation openings clear where applicable.
2. Connect the stove to the approved electrical supply without overloading sockets or using damaged extension leads.
3. Place the filled cookware centrally on the designated heating zone before switching on where the model requires it.
4. Switch on the stove and select a suitable heat setting for the cooking task.
5. Allow the cookware to heat gradually and monitor the food to prevent overheating or boiling over.
6. Adjust the heat as required and use suitable utensils to stir or handle the food.

7. Keep pot handles turned inward where safe and avoid leaving hot cookware near the edge of the surface.
8. If liquid spills onto the stove, switch it off and disconnect power if safe to do so; allow the unit to cool before cleaning.
9. If the stove emits smoke, sparks, an unusual smell, or trips the electrical protection, switch it off and stop using it.
10. When cooking is complete, switch the controls off and disconnect the power if required by the manufacturer's instructions.
11. Treat the cooking surface and cookware as hot until fully cooled; use suitable heat-resistant gloves when needed.
12. After cooling, clean food spills using the method approved for the stove type and avoid wetting electrical parts.
13. Store the stove in a dry location and inspect it before the next use.

⚠ Safety

- Use the stove according to the manufacturer's instructions and its rated electrical supply.
- Never operate the stove with damaged wiring, plugs, controls, or a cracked cooking surface.
- Keep water and wet hands away from plugs, sockets, and electrical controls.
- Do not leave cooking unattended when heat, oil, or combustible materials are present.
- Keep flammable items, paper, and loose cloth away from hot surfaces.
- Do not touch heating elements or hot cookware with bare hands; allow sufficient cooling time.
- Do not immerse the appliance or its electrical components in water unless the manufacturer explicitly permits it.
- If an electrical fire occurs, disconnect power only if safe and never throw water onto energised electrical equipment or burning oil.

Care & storage

- Switch off, unplug where applicable, and allow the stove to cool before cleaning.
- Remove food residue using a soft cloth and a cleaning method suitable for the surface type.
- Inspect the power cord, plug, heating element, and controls regularly for wear or damage.
- Keep ventilation openings clear and avoid abrasive tools that may damage the cooking surface.
- Arrange qualified servicing if heating becomes uneven, controls fail, or electrical protection trips repeatedly.

Common mistakes to avoid

- Using cookware unsuitable for the stove type or too small for the heating zone.
- Operating the stove near water or with damaged electrical connections.
- Leaving the stove unattended or setting heat higher than necessary.
- Cleaning the appliance while hot or allowing water to enter electrical components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.